

LOW POWER QUAD COMPARATOR

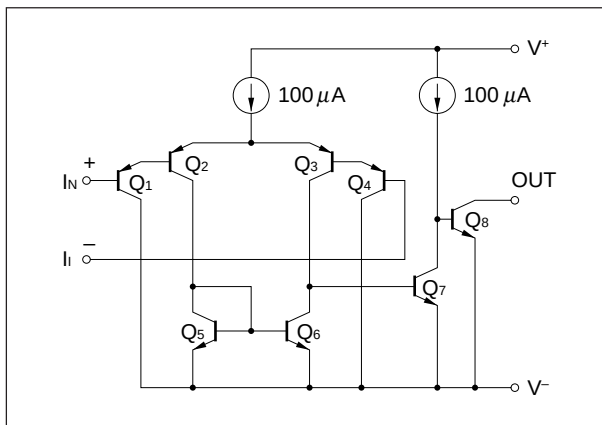
DESCRIPTION

The μ PC339 is a quad comparator which is designed to operate from a single power supply over a wide range of voltage. Operation from split power supplies, is also possible and the power supply current drain is very low. Further advantage, the input common-mode voltage includes ground, even though operated from a single power supply voltage.

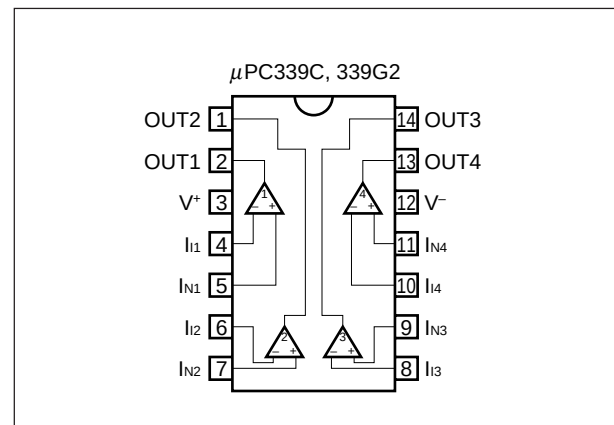
FEATURES

- Common-mode input voltage range includes V^-
- Wide supply voltage range
2 V to 32 V (Single)
 ± 1 V to ± 16 V (Split)
- Low supply current
- Open collector output

EQUIVALENT CIRCUIT (1/4 Circuit)



PIN CONFIGURATION (Top View)



ORDERING INFORMATION

Part Number	Package
μ PC339C	14-pin plastic DIP (300 mil)
μ PC339G2	14-pin plastic SOP (225 mil)

The information in this document is subject to change without notice.

ABSOLUTE MAXIMUM RATINGS ($T_A = 25\text{ }^{\circ}\text{C}$)

Parameter		Symbol	Ratings	Unit
Voltage between V^+ and V^-	Note 1	$V^+ - V^-$	-0.3 to +36	V
Differential Input Voltage		V_{ID}	± 36	V
Input Voltage	Note 2	V_I	$V^- - 0.3$ to $V^- + 36$	V
Output Voltage	Note 3	V_O	$V^- - 0.3$ to $V^- + 36$	V
Power Dissipation	C Package Note 4	P_T	570	mW
	G2 Package Note 5		550	mW
Output Short Circuit Duration	Note 6		Indefinite	sec
Operating Ambient Temperature		T_A	-20 to +80	$^{\circ}\text{C}$
Storage Temperature		T_{stg}	-55 to +125	$^{\circ}\text{C}$

Notes 1. Reverse connection of supply voltage can cause destruction.

2. The input voltage should be allowed to input without damage or destruction independent of the magnitude of V^+ . Either input signal should not be allowed to go negative by more than 0.3 V. The normal operation will establish when any input is within the Common Mode Input Voltage Range of electrical characteristics.
3. This specification is the voltage which should be allowed to supply to the output terminal from external without damage or destruction independent of the magnitude of V^+ . Even during the transition period of supply voltage, power on/off etc., this specification should be kept.
4. Thermal derating factor is $-7.6\text{ mW}/^{\circ}\text{C}$ when operating ambient temperature is higher than $55\text{ }^{\circ}\text{C}$.
5. Thermal derating factor is $-5.5\text{ mW}/^{\circ}\text{C}$ when operating ambient temperature is higher than $25\text{ }^{\circ}\text{C}$.
6. Short circuits from the output to V^+ can cause destruction. Pay careful attention to the total power dissipation not to exceed the absolute maximum ratings, Note 4 and Note 5.

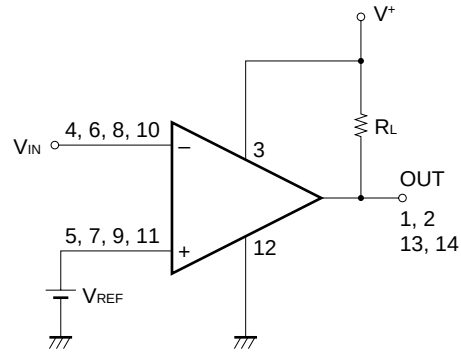
RECOMMENDED OPERATING CONDITIONS

Parameter	Symbol	MIN.	TYP.	MAX.	Unit
Supply Voltage (Split)	$V^{\pm}$	± 1		± 16	V
Supply Voltage ($V^- = \text{GND}$)	V^+	+2		+32	V

ELECTRICAL CHARACTERISTICS ($T_A = 25\text{ }^{\circ}\text{C}$, $V^+ = 5\text{ V}$, $V^- = \text{GND}$)

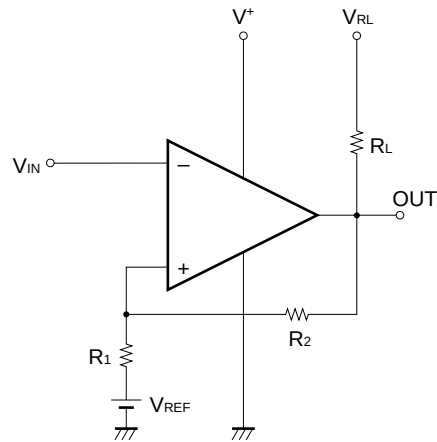
Parameter	Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Input Offset Voltage	V_{IO}	$V_O = 1.4\text{ V}$, $V_{REF} = 1.4\text{ V}$, $R_S = 0\text{ }\Omega$		± 2	± 5	mV
Input Offset Current	I_{IO}	$V_O \doteq 1.4\text{ V}$		± 5	± 50	nA
Input Bias Current	I_B	$V_O \doteq 1.4\text{ V}$		25	250	nA
Voltage Gain	A_V	$R_L = 15\text{ k}\Omega$		200		V/mV
Supply Current	I_{CC}	$R_L = \infty$, $I_O = 0\text{ A}$, All Comparators		0.8	2	mA
Common Mode Input Voltage Range	V_{ICM}		0		$V^+ - 1.5$	V
Output Saturation Voltage	V_{OL}	$V_{IN(-)} = 1\text{ V}$, $V_{IN(+)} = 0\text{ V}$, $I_{O\text{ SINK}} = 4\text{ mA}$		0.2	0.4	V
Output Sink Current	$I_{O\text{ SINK}}$	$V_{IN(-)} = 1\text{ V}$, $V_{IN(+)} = 0\text{ V}$, $V_O \leq 1.5\text{ V}$	6	16		mA
Output Leakage Current	$I_{O\text{ LEAK}}$	$V_{IN(+)} = 1\text{ V}$, $V_{IN(-)} = 0\text{ V}$, $V_O = 5\text{ V}$		0.1		nA
Response Time		$R_L = 5.1\text{ k}\Omega$, $V_{RL} = 5\text{ V}$		1.3		μs

APPLICATION CIRCUIT EXAMPLE



V_{REF} : V^- to $V^+ - 1.5$ (V)

COMPARATOR with HYSTERESIS CIRCUIT

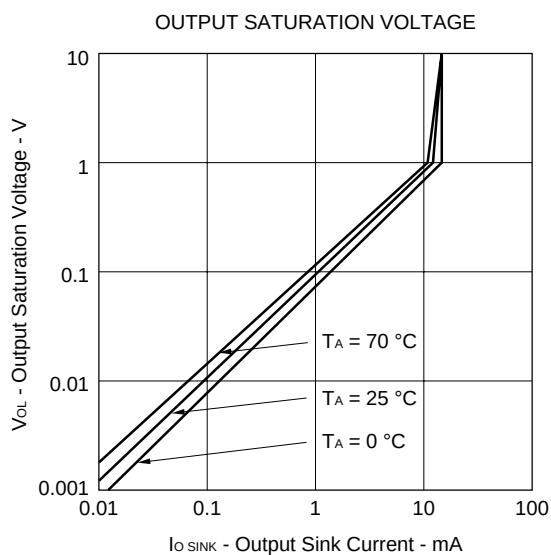
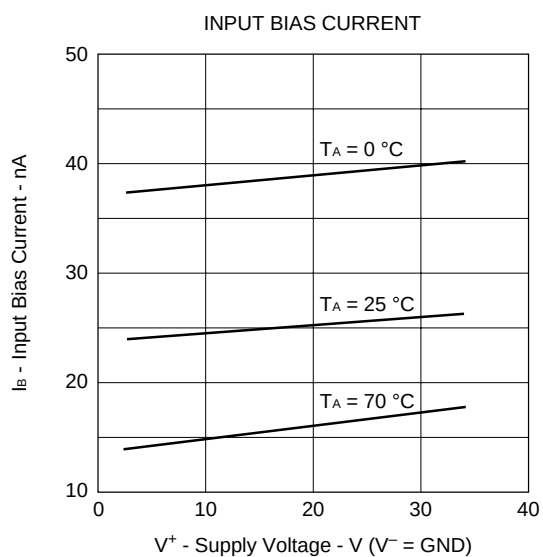
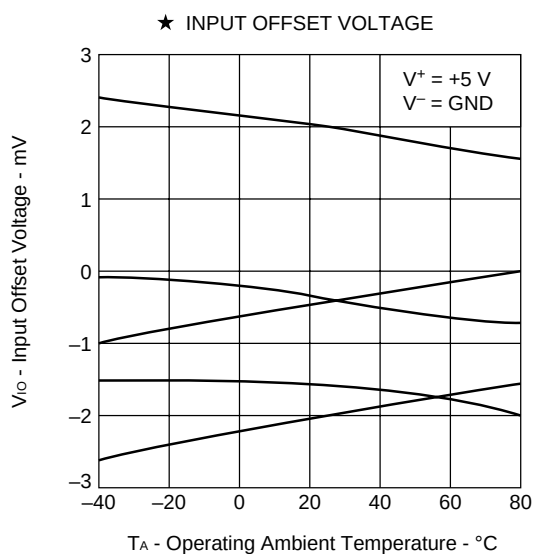
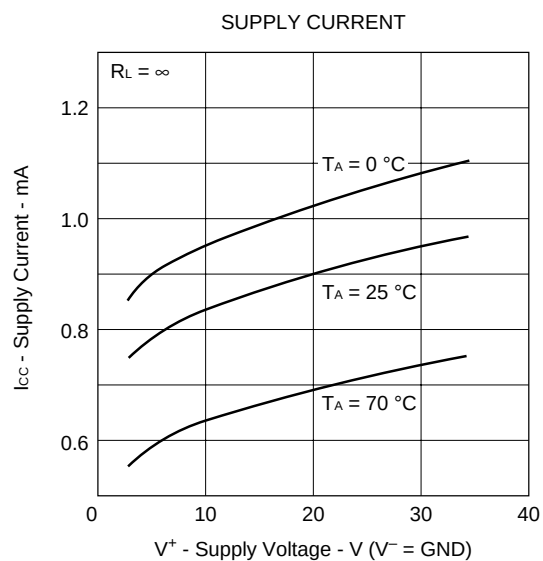
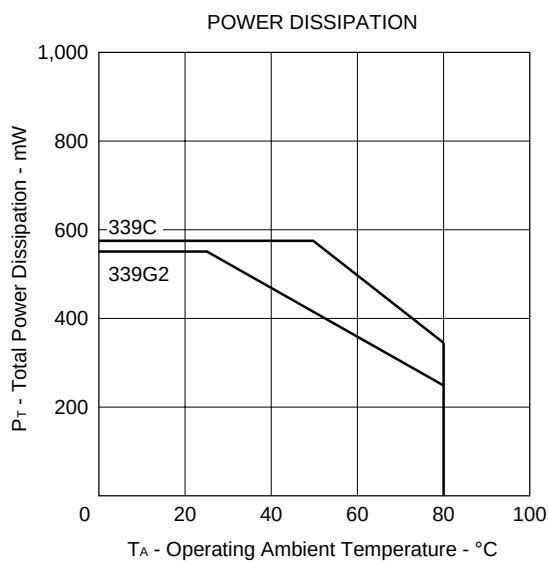


- Threshold voltage

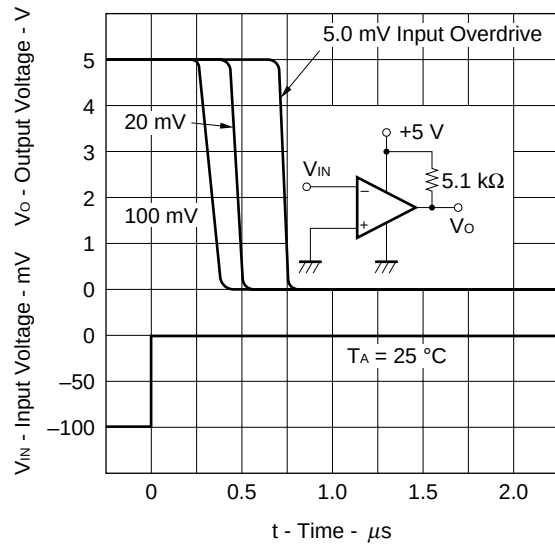
$$V_{TH(High)} \doteq V_{REF} + \frac{R_1}{R_L + R_2 + R_1} (V_{RL} - V_{REF})$$

$$V_{TH(Low)} \doteq V_{REF} - \frac{R_1}{R_1 + R_2} (V_{REF} - V_{OL})$$

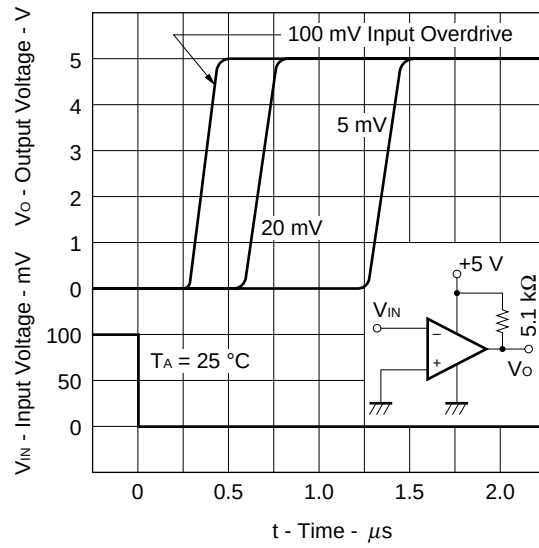
$$(V_{RL} > V_{REF} > V_{OL})$$

TYPICAL PERFORMANCE CHARACTERISTICS ($T_A = 25\text{ }^\circ\text{C}$, TYP.)

RESPONSE TIME FOR VARIOUS INPUT OVERDRIVES I

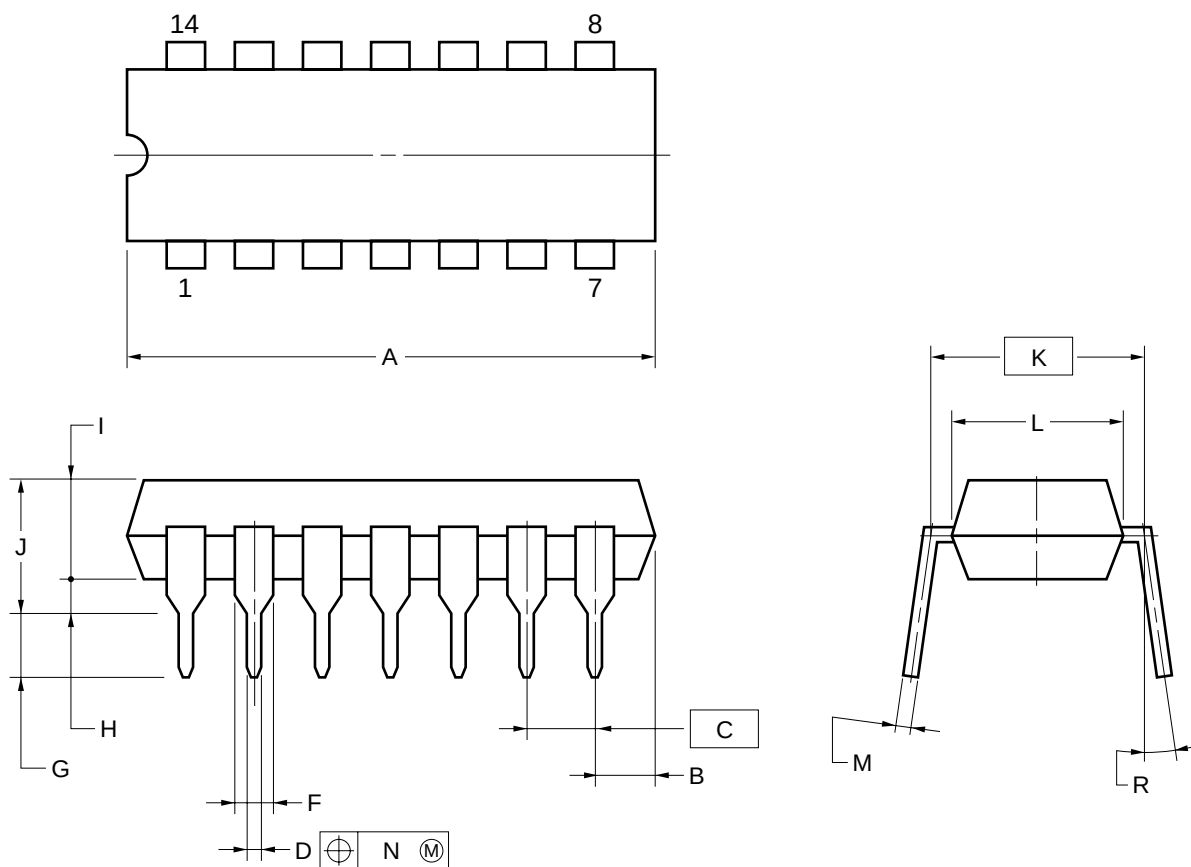


RESPONSE TIME FOR VARIOUS INPUT OVERDRIVES II



PACKAGE DRAWINGS

14PIN PLASTIC DIP (300 mil)



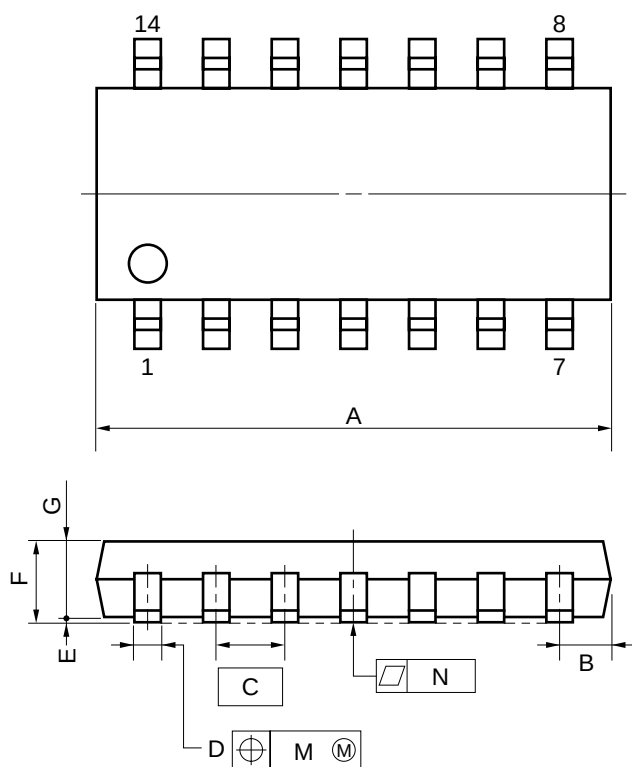
NOTES

- 1) Each lead centerline is located within 0.25 mm (0.01 inch) of its true position (T.P.) at maximum material condition.
- 2) Item "K" to center of leads when formed parallel.

ITEM	MILLIMETERS	INCHES
A	20.32 MAX.	0.800 MAX.
B	2.54 MAX.	0.100 MAX.
C	2.54 (T.P.)	0.100 (T.P.)
D	0.50±0.10	0.020 ^{+0.004} _{-0.005}
F	1.2 MIN.	0.047 MIN.
G	3.6±0.3	0.142±0.012
H	0.51 MIN.	0.020 MIN.
I	4.31 MAX.	0.170 MAX.
J	5.08 MAX.	0.200 MAX.
K	7.62 (T.P.)	0.300 (T.P.)
L	6.4	0.252
M	0.25 ^{+0.10} _{-0.05}	0.010 ^{+0.004} _{-0.003}
N	0.25	0.01
R	0~15°	0~15°

P14C-100-300B1-1

14 PIN PLASTIC SOP (225 mil)



NOTE

Each lead centerline is located within 0.12 mm (0.005 inch) of its true position (T.P.) at maximum material condition.

ITEM	MILLIMETERS	INCHES
A	10.46 MAX.	0.412 MAX.
B	1.42 MAX.	0.056 MAX.
C	1.27 (T.P.)	0.050 (T.P.)
D	$0.40^{+0.10}_{-0.05}$	$0.016^{+0.004}_{-0.003}$
E	0.1 ± 0.1	0.004 ± 0.004
F	1.8 MAX.	0.071 MAX.
G	1.49	0.059
H	6.5 ± 0.3	0.256 ± 0.012
I	4.4	0.173
J	1.1	0.043
K	$0.15^{+0.10}_{-0.05}$	$0.006^{+0.004}_{-0.002}$
L	0.6 ± 0.2	$0.024^{+0.008}_{-0.009}$
M	0.12	0.005
N	0.10	0.004
P	$3^{\circ} \text{ } ^{+7^{\circ}}_{-3^{\circ}}$	$3^{\circ} \text{ } ^{+7^{\circ}}_{-3^{\circ}}$

S14GM-50-225B, C-4

RECOMMENDED SOLDERING CONDITIONS

When soldering this product, it is highly recommended to observe the conditions as shown below. If other soldering processes are used, or if the soldering is performed under different conditions, please make sure to consult with our sales offices.

For more details, refer to our document “**SEMICONDUCTOR DEVICE MOUNTING TECHNOLOGY MANUAL**” (C10535E).

Surface mount device

μPC339G2: 14-pin plastic SOP (225 mil)

Process	Conditions	Symbol
Infrared ray reflow	Peak temperature: 230 °C or below (Package surface temperature), Reflow time: 30 seconds or less (at 210 °C or higher), Maximum number of reflow processes: 1 time.	IR30-00-1
Vapor Phase Soldering	Peak temperature: 215 °C or below (Package surface temperature), Reflow time: 40 seconds or less (at 200 °C or higher), Maximum number of reflow processes: 1 time.	VP15-00-1
Wave Soldering	Solder temperature: 260 °C or below, Flow time: 10 seconds or less, Maximum number of flow processes: 1 time, Pre-heating temperature: 120 °C or below (Package surface temperature).	WS60-00-1
Partial heating method	Pin temperature: 300 °C or below, Heat time: 3 seconds or less (Per each side of the device).	—

Caution Apply only one kind of soldering condition to a device, except for “partial heating method”, or the device will be damaged by heat stress.

Through-hole device

μPC339C: 14-pin plastic DIP (300 mil)

Process	Conditions
Wave soldering (only to leads)	Solder temperature: 260 °C or below, Flow time: 10 seconds or less.
Partial heating method	Pin temperature: 300 °C or below, Heat time: 3 seconds or less (per each lead.)

Caution For through-hole device, the wave soldering process must be applied only to leads, and make sure that the package body does not get jet soldered.

REFERENCE DOCUMENTS

QUALITY GRADES ON NEC SEMICONDUCTOR DEVICES	C11531E
SEMICONDUCTOR DEVICE MOUNTING TECHNOLOGY MANUAL	C10535E
IC PACKAGE MANUAL	C10943X
GUIDE TO QUALITY ASSUARANCE FOR SEMICONDUCTOR DEVICES	MEI-1202
SEMICONDUCTORS SELECTION GUIDE	X10679E
NEC SEMICONDUCTOR DEVICE RELIABILITY/	IEI-1212
QUALITY CONTROL SYSTEM - STANDARD LINEAR IC	

[MEMO]

[MEMO]

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Standard: Computers, office equipment, communications equipment, test and measurement equipment, audio and visual equipment, home electronic appliances, machine tools, personal electronic equipment and industrial robots

Special: Transportation equipment (automobiles, trains, ships, etc.), traffic control systems, anti-disaster systems, anti-crime systems, safety equipment and medical equipment (not specifically designed for life support)

Specific: Aircrafts, aerospace equipment, submersible repeaters, nuclear reactor control systems, life support systems or medical equipment for life support, etc.

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Anti-radioactive design is not implemented in this product.